



# Technical Service Bulletin

Technical Service Bulletin: tsb101155

Released Date: 13-Sep-2017

Integrated Fuel System Module (IFSM) Rebuild Instructions

## Integrated Fuel System Module (IFSM) Rebuild Instructions

### Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

### Contents

#### Product Affected

- ISX CM570
- ISX CM870
- ISX CM871
- QSX CM570

This Service/Parts Topic introduces a rebuild kit designed to assist the rebuild of the integrated fuel system module (IFSM) and also updates procedure 005-073 in the Signature, ISX, QSX Troubleshooting and Repair Manual, Bulletin 3666239.

The tables shown below summarize the part numbers of the contents included in the Rebuild Kit, as well as reusable components.

**Table 1, Integrated Fuel System Module (IFSM) Rebuild Kit Contents, Part Number 4025255, for CM570**

| Part Number | Housing Assembly Content | Description                                 | Quantity |
|-------------|--------------------------|---------------------------------------------|----------|
| S-2286      | -                        | Screw, drive                                | 2        |
| 105375      | -                        | Nameplate                                   | 1        |
| 129888      | -                        | Seal, rectangular ring (fuel shutoff valve) | 2        |
| 151900      | -                        | Seal, rectangular ring (damper)             | 2        |
| 3045986     | -                        | Seal, o-ring (actuator)                     | 4        |
| 3089240     | -                        | Seal, o-ring (check valve)                  | 2        |

**Table 1, Integrated Fuel System Module (IFSM) Rebuild Kit Contents, Part Number 4025255, for CM570**

| <b>Part Number</b> | <b>Housing Assembly Content</b> | <b>Description</b>                                    | <b>Quantity</b> |
|--------------------|---------------------------------|-------------------------------------------------------|-----------------|
| 3330478            | -                               | Seal, o-ring (2620 kPa regulator) [380 psi regulator] | 1               |
| 3348703            | -                               | Seal, o-ring (1724 kPa regulator) [250 psi regulator] | 1               |
| 3678786            | -                               | Seal, o-ring (male union)                             | 3               |
| 3678912            | -                               | Seal, o-ring (M14 plugs and Compuchek® fitting)       | 6               |
| 3679139            | -                               | Seal, o-ring (drain fitting)                          | 1               |
| 3867481            | -                               | Seal, o-ring (fuel pump)                              | 1               |
| 3867640            | -                               | Seal, o-ring (1724 kPa regulator) [250 psi regulator] | 14              |
| 3867641            | -                               | Seal, o-ring (2620 kPa regulator) [380 psi regulator] | 1               |
| 4010519            | -                               | Seal, o-ring (temperature and pressure sensors)       | 3               |
| 2872195            | -                               | Gasket, support (fuel pump)                           | 1               |
| 4965689            | -                               | Gasket, fuel control housing                          | 1               |
| 4076597            | -                               | Housing, fuel control assembly                        | 1               |
| -                  | 4984297                         | Housing, fuel control<br>Housing, fuel control        | 1               |
| -                  | 3015521                         | Ball                                                  | 1               |
| -                  | 3330019                         | Screw, socket head capscREW                           | 1               |
| -                  | 3330719                         | Plate, thrust bearing                                 | 1               |
| -                  | 3330720                         | Shaft, idler                                          | 1               |

**Table 1, Integrated Fuel System Module (IFSM) Rebuild Kit Contents, Part Number 4025255, for CM570**

| <b>Part Number</b> | <b>Housing Assembly Content</b> | <b>Description</b>            | <b>Quantity</b> |
|--------------------|---------------------------------|-------------------------------|-----------------|
| -                  | 3331275                         | Retainer, shaft               | 1               |
| -                  | 3867093                         | Gear, idler                   | 1               |
| -                  | 4076593                         | Valve, air bleed check        | 1               |
| -                  | 4076599                         | Plug, orifice                 | 1               |
| 4954473            | -                               | Gasket, fuel pump (lift pump) | 1               |
| 3025806            | -                               | Screw, socket head cap        | 3               |
| 3021123            | -                               | Seal, o-ring                  | 1               |
| 3922794            | -                               | Seal, o-ring                  | 2               |
| 3037537            | -                               | Seal, o-ring                  | 2               |
| 3042543            | -                               | Seal, o-ring                  | 1               |
| 3089392            | -                               | Seal, o-ring                  | 1               |
| 3084539            | -                               | Seal, o-ring                  | 1               |
| 3678846            | -                               | Seal, o-ring                  | 2               |
| 4076585            | -                               | Plate, cover                  | 1               |
| 3347832            | -                               | Seal, grommet                 | 4               |
| 3918190            | -                               | Washer, sealing               | 2               |

**Table 2, Integrated Fuel System Module (IFSM) Kit Content, Part Number 4955685, for CM870 and CM871**

| <b>Part Number</b> | <b>Housing Assembly Content</b> | <b>Description</b>                          | <b>Quantity</b> |
|--------------------|---------------------------------|---------------------------------------------|-----------------|
| S-2286             | -                               | Screw, drive                                | 2               |
| 105375             | -                               | Nameplate                                   | 1               |
| 129888             | -                               | Seal, rectangular ring (fuel shutoff valve) | 2               |
| 151900             | -                               | Seal, rectangular ring (damper)             | 2               |
| 3045986            | -                               | Seal, o-ring (actuator)                     | 4               |
| 3089240            | -                               | Seal, o-ring (check valve)                  | 2               |

**Table 2, Integrated Fuel System Module (IFSM) Kit Content, Part Number 4955685, for CM870 and CM871**

| <b>Part Number</b> | <b>Housing Assembly Content</b> | <b>Description</b>                                    | <b>Quantity</b> |
|--------------------|---------------------------------|-------------------------------------------------------|-----------------|
| 3330478            | -                               | Seal, o-ring (2620 kPa regulator) [380 psi regulator] | 1               |
| 3348703            | -                               | Seal, o-ring (1724 kPa regulator) [250 psi regulator] | 1               |
| 3627695            | -                               | Seal, o-ring (male union)                             | 2               |
| 3678786            | -                               | Seal, o-ring (male union)                             | 2               |
| 3678912            | -                               | Seal, o-ring (M14 plugs and Compuchek® fitting)       | 7               |
| 3679139            | -                               | Seal, o-ring (drain fitting)                          | 1               |
| 3867481            | -                               | Seal, o-ring (fuel pump)                              | 1               |
| 3867640            | -                               | Seal, o-ring (1724 kPa regulator) [250 psi regulator] | 13              |
| 3867641            | -                               | Seal, o-ring (2620 kPa regulator) [380 psi regulator] | 1               |
| 4010519            | -                               | Seal, o-ring (temperature and pressure sensors)       | 2               |
| 2872195            | -                               | Gasket, support (fuel pump)                           | 1               |
| 4965689            | -                               | Gasket, fuel control housing                          | 1               |
| 4921791            | -                               | Housing, fuel control assembly                        | 1               |
| -                  | 4984296                         | Housing, fuel controlHousing, fuel control            | 1               |
| -                  | 3015521                         | Ball                                                  | 1               |
| -                  | 3330019                         | Screw, socket head capscrew                           | 1               |

**Table 2, Integrated Fuel System Module (IFSM) Kit Content, Part Number 4955685, for CM870 and CM871**

| Part Number | Housing Assembly Content | Description                   | Quantity |
|-------------|--------------------------|-------------------------------|----------|
| -           | 3330719                  | Plate, thrust bearing         | 1        |
| -           | 3330720                  | Shaft, idler                  | 1        |
| -           | 3331275                  | Retainer, shaft               | 1        |
| -           | 3867093                  | Gear, idler                   | 1        |
| -           | 4076593                  | Valve, air bleed check        | 1        |
| -           | 4076599                  | Plug, orifice                 | 1        |
| 4954474     | -                        | Gasket, fuel pump (lift pump) | 1        |
| 4088806     | -                        | Gasket, fuel pump             | 1        |

**Table 3, Reusable Components**

| Part Number | Description                                                        | Quantity |
|-------------|--------------------------------------------------------------------|----------|
| 70704       | Washer, plain (gear pump mounting)                                 | 4        |
| 129768      | Spring, valve (FSO assembly)                                       | 1        |
| 129839      | Shield, shutoff valve                                              | 1        |
| 196057      | Disc, valve (FSO assembly)                                         | 1        |
| 3025806     | Screw, socket head capscrew (lift pump mounting)                   | 4        |
| 4088863     | Housing, actuator (FSO assembly)                                   | 1        |
| 4088274     | Disc, actuator (FSO assembly)                                      | 1        |
| 3100221     | Nipple, coupling (Compuchek® fitting)                              | 2        |
| 4076472     | Screw, socket head capscrew (actuator mounting)                    | 12       |
| 4076471     | Screw, socket head cap (FSO mounting)                              | 4        |
| 3331873     | Union, male                                                        | 1        |
| 3348322     | Valve, pressure regulator (1724 kPa regulator) [250 psi regulator] | 1        |

| Table 3, Reusable Components           |                                                                        |          |
|----------------------------------------|------------------------------------------------------------------------|----------|
| Part Number                            | Description                                                            | Quantity |
| 3397506                                | Screw, socket head capscrew (fuel damper mounting)                     | 4        |
| 3408421                                | Solenoid (FSO assembly)                                                | 1        |
| 3417138                                | Elbow, male adapter (fuel drain line number 8)                         | 1        |
| 3678921                                | Plug, threaded                                                         | 5        |
| 3681694                                | Fitting, inlet                                                         | 1        |
| 4954905                                | Sensor, temperature                                                    | 1        |
| 3902023                                | Screw, hexagon flange head capscrew (fuel damper assembly)             | 2        |
| 4009956                                | Valve, pressure regulator (2620 kPa regulator) [380 psi regulator]     | 1        |
| 4010077                                | Valve, check                                                           | 1        |
| 3678873                                | Plug, threaded                                                         | 1        |
| 4061815                                | Actuator, metering                                                     | 2        |
| 4061816                                | Actuator, timing                                                       | 2        |
| 4088847                                | Pump, gear fuel                                                        | 1        |
| 4076858                                | Screw, socket head capscrew (gear pump mounting)                       | 4        |
| 4921519                                | Sensor, pressure                                                       | 1        |
| 4076857 (+12-VDC) or 4935093 (+24-VDC) | Pump, fuel (lift pump) - EGR differential pressure sensor side mounted | 1        |
| 5362253 (+12-VDC or 4935093 (+24-VDC)  | Pump, fuel (lift pump) - EGR differential pressure sensor rear mounted | 1        |
| -                                      | Damper, fuel control (assembly)                                        | 2        |

| Table 4, Integrated Fuel System Module (IFSM) Configuration Chart |                           |         |         |         |         |
|-------------------------------------------------------------------|---------------------------|---------|---------|---------|---------|
|                                                                   |                           | 4076844 | 4077280 | 4088505 | 4088510 |
| Electric Lift Pump                                                | Yes +12 VDC solenoid pump |         |         |         |         |

**Table 4, Integrated Fuel System Module (IFSM) Configuration Chart**

|                                  |                                    | <b>4076844</b> | <b>4077280</b> | <b>4088505</b> | <b>4088510</b> |
|----------------------------------|------------------------------------|----------------|----------------|----------------|----------------|
| Electric Lift Pump               | Yes +24 VDC solenoid pump          |                |                |                |                |
| Electric Lift Pump               | Yes +12 VDC engine mount vane pump |                |                |                |                |
| Electric Lift Pump               | Yes +24 VDC engine mount vane pump |                |                |                |                |
| Electric Lift Pump               | Yes +12 VDC IFSM mount vane pump   | X              |                | X              |                |
| Electric Lift Pump               | Yes +24 VDC IFSM mount vane pump   |                | X              |                | X              |
| Electric Lift Pump               | No                                 |                |                |                |                |
| EGR Differential Pressure Sensor | Mounted to side of lift pump       | X              | X              |                |                |
| EGR Differential Pressure Sensor | Mounted to rear of lift pump       |                |                | X              | X              |

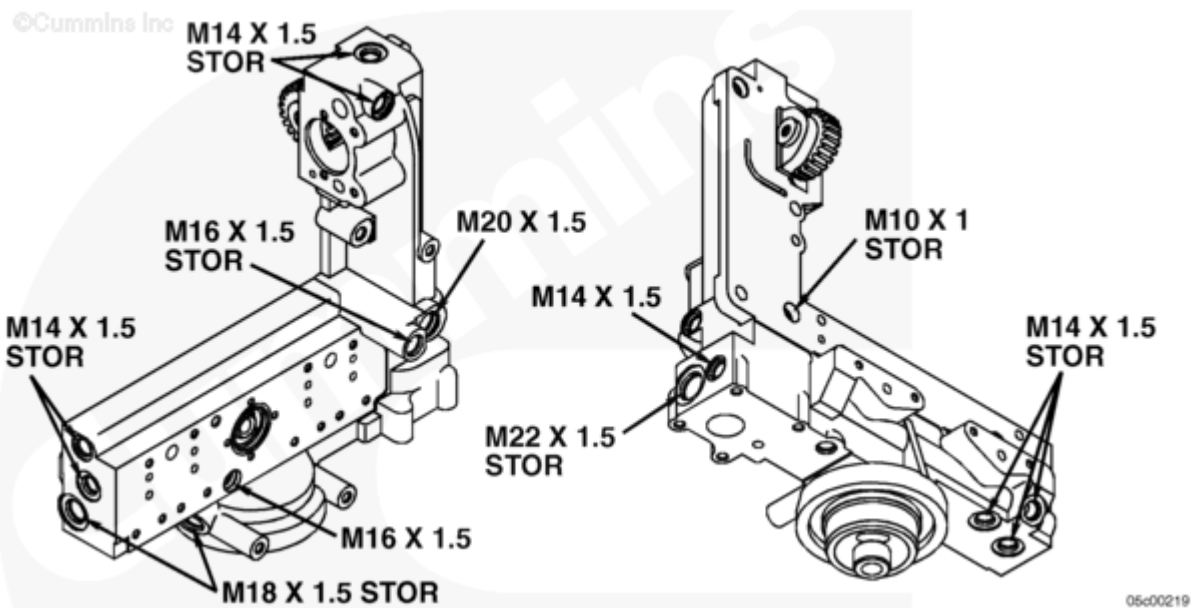


Figure 1, Integrated Fuel System Module (IFSM) Port Locations.

### Related Procedures

Refer to Procedure 005-073 (</qs3/pubsys2/xml/en/procedures/10/10-005-073-tr.html>) Integrated Fuel System Module (IFSM).

## Document History

| Date      | Details                                                  |
|-----------|----------------------------------------------------------|
| 2015-3-26 | Broken link                                              |
| 2017-6-19 | Updated Table 1 and added Table 2.                       |
| 2017-9-12 | Added Product Affected Section and updated part numbers. |

Last Modified: 13-Sep-2017